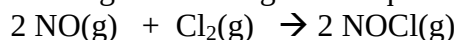


CHAPTER 14
Chemical Kinetics

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1. Which of the following expressions correctly describes the relationship between the rates at which NO_2 and Cl_2 are consumed in the reaction below? $2 \text{NO}_2(\text{g}) + \text{Cl}_2(\text{g}) \rightarrow 2 \text{NO}_2\text{Cl}(\text{g})$
- A. $\frac{-\Delta(\text{NO}_2)}{\Delta t} = -\frac{1}{2} \left[\frac{\Delta(\text{Cl}_2)}{\Delta t} \right]$ B. $\frac{-\Delta(\text{NO}_2)}{\Delta t} = -2 \left[\frac{\Delta(\text{Cl}_2)}{\Delta t} \right]$
- C. $\frac{-\Delta(\text{NO}_2)}{\Delta t} = \frac{1}{2} \left[\frac{\Delta(\text{Cl}_2)}{\Delta t} \right]$ D. $-\frac{\Delta(\text{NO}_2)}{\Delta t} = -2 \left[\frac{\Delta(\text{Cl}_2)}{\Delta t} \right]$
-
2. A particular chemical reaction involves a single reactant. What is the order of the reaction if the rate increases by a factor of eight when the concentration of the reactant is doubled?
- A. zero order B. 1st order C. 2nd order D. 3rd order
-
3. Gaseous N_2O_5 decomposes according to the following equation:
- $$\text{N}_2\text{O}_5(\text{g}) \rightarrow 2 \text{NO}_2(\text{g}) + 1/2 \text{O}_2(\text{g})$$
- The experimental rate law is $-\Delta[\text{N}_2\text{O}_5]/\Delta t = k[\text{N}_2\text{O}_5]$. At a certain temperature the rate constant is $k = 5.0 \times 10^{-4}/\text{second}$. In how many seconds will the concentration of N_2O_5 decrease to one-tenth of its initial value?
- A. $2.0 \times 10^3 \text{ s}$ B. $4.6 \times 10^3 \text{ s}$ C. $2.1 \times 10^2 \text{ s}$ D. $1.4 \times 10^3 \text{ s}$
-
4. If a reaction is zero order in a reactant, when the concentration of the reactant is decreased by a factor of 2, the reaction rate will
- A. decrease by a factor of $1/4$ B. remain constant.
- C. decrease by a factor of $1/2$ D. double
-
5. At 350 K, a particular second-order reaction, consisting of a single reactant, **A**, has a rate constant equal to $4.5 \times 10^{-3} \text{ M}^{-1} \text{ s}^{-1}$. If the initial concentration of **A** is 0.80 M, how many half-lives are required for the concentration of **A** to become equal to 0.10 M?
- A. 1 B. 2 C. 3 D. 4
-
6. Consider the following chemical reaction:
- $$2 \text{H}_2\text{O}_2(\text{aq}) \rightarrow 2 \text{H}_2\text{O}(\text{l}) + \text{O}_2(\text{g})$$
- If the average rate of disappearance of H_2O_2 over a certain time interval is $6.80 \times 10^{-5} \text{ M s}^{-1}$, what is the average rate of appearance of O_2 during this same time interval?
- A. $4.62 \times 10^{-9} \text{ M s}^{-1}$ B. $3.40 \times 10^{-5} \text{ M s}^{-1}$ C. $6.80 \times 10^{-5} \text{ M s}^{-1}$ D. $1.36 \times 10^{-4} \text{ M s}^{-1}$
-
7. Consider the following reaction:
- $$2 \text{NOBr}(\text{g}) \rightarrow 2 \text{NO}(\text{g}) + \text{Br}_2(\text{g})$$
- The reaction is known to be second order with $k = 0.80 \text{ L/mol s}$. If you start with a concentration of 0.086 mol/L of NOBr , what will be its concentration after 22 seconds?
- A. $2.0 \times 10^{-9} \text{ mol/L}$ B. $5.4 \times 10^{-2} \text{ mol/L}$ C. $3.4 \times 10^{-2} \text{ mol/L}$ D. $8.7 \times 10^{-1} \text{ mol/L}$
-
8. The rate laws for certain enzyme-activated reactions in your body have a specific rate constant k , with units of M/s . What is the overall order of these reactions?
- A. 1 B. 2 C. 3 D. 0
-
9. The rate constant of a first-order reaction is $4.60 \times 10^{-4}/\text{s}$ at 350°C . If the activation energy is 104 kJ/mol , at what temperature will the rate constant be $8.80 \times 10^{-4}/\text{s}$?
- A. 669°C B. 402°C C. 371°C D. 350°C
-

Q10-Q12: Use the following data for the next three questions:

Nitric oxide gas reacts with chlorine gas according to the equation,



The following data were obtained for this reaction:

initial [NO] M	initial [Cl ₂] M	initial rate mol/L hr
0.50	0.50	1.19
1.00	0.50	4.79
1.00	1.00	9.59
1.50	1.50	32.27

10. Which of the following rate laws is consistent with these data?

A. Rate = $k[\text{NO}]$

B. Rate = $k[\text{NO}][\text{Cl}_2]^{1/2}$

C. Rate = $k[\text{NO}][\text{Cl}_2]$

D. Rate = $k[\text{NO}]^2[\text{Cl}_2]$

11. What is the **OVERALL** order of the reaction of NO with Cl₂?

A. 1

B. 1.5

C. 3

D. 22

12. What is the value of the rate constant for the reaction of NO with Cl₂?

A. 2.38

B. 3.35

C. 4.76

D. 9.52

13. In terms of the "Collision Theory of Chemical Kinetics", the rate of a chemical reaction is proportional to:

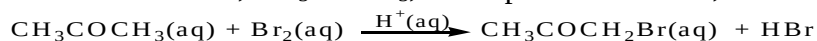
A. the change in free energy per second.

B. the change in temperature per second.

C. the number of collisions per second.

D. the number of product molecules.

14. Consider the bromination of acetone, CH₃COCH₃, in the presence of acid,



and the experimental data shown below

[CH ₃ COCH ₃], M	[Br ₂], M	[H ⁺], M	initial rate, M s ⁻¹
0.30	0.050	0.050	5.70×10^{-5}
0.30	0.10	0.050	5.70×10^{-5}
0.30	0.050	0.10	1.14×10^{-4}
0.60	0.050	0.20	4.56×10^{-4}
0.60	0.050	0.050	1.14×10^{-4}

Which of the following is the correct rate law for this reaction?

A. Rate = $k[\text{CH}_3\text{COCH}_3][\text{Br}_2][\text{H}^+]$

B. Rate = $k[\text{CH}_3\text{COCH}_3]$

C. Rate = $k[\text{CH}_3\text{COCH}_3][\text{Br}_2]$

D. Rate = $k[\text{CH}_3\text{COCH}_3][\text{H}^+]$

15. Which of the following would be expected to affect the rate of a given chemical reaction?

I. The reaction temperature.

II. The concentration of the reactants.

III. A catalyst

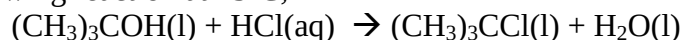
A. I

B. II

C. I and III only

D. I, II, and III

16. Consider the following reaction at 25°C,



The experimentally determined rate law for this reaction indicates that the reaction is first-order in (CH₃)₃COH and that the reaction is first-order overall. Which of the following would produce an increase in the rate of this reaction?

A. increasing the concentration of (CH₃)₃COH

B. increasing the concentration of HCl

C. decreasing the concentration of (CH₃)₃CCl

D. decreasing the concentration of HCl

17. A student determined the value of the rate constant, k , for a chemical reaction at several different temperatures. Which of the following graphs of the student's data would give a straight line?

- A. k versus T B. k versus $(1/T)$ C. $\ln k$ versus $(1/T)$ D. $\ln k$ versus T
-

18. Which of the following would **DECREASE** the rate of a chemical reaction?

- A. decreasing the activation energy.
B. increasing the concentrations of the reactants.
C. increasing the temperature.
D. adding a catalyst.
E. None of these will decrease the rate.
-

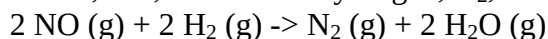
19. Which of the following statements is **TRUE**? For a fast reaction, the half-life.

- A. the values of the rate constant and the half-life are both large.
B. the values of the rate constant and the half-life are both small.
C. the value of the rate constant is small and the value of the half-life is large.
D. the value of the rate constant is large and the value of the half-life is small.
-

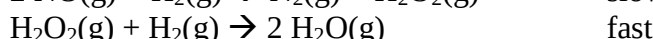
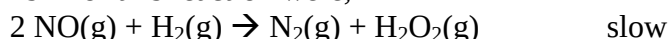
20. Which of the following statements best describes how a catalyst works?

- A. A catalyst changes the potential energies of the reactants and products.
B. A catalyst decreases the temperature of the reaction which leads to a faster rate.
C. A catalyst lowers the activation energy for the reaction by providing a different reaction mechanism.
D. A catalyst raises the activation energy for the reaction which produces a faster rate.
-

21. Nitrogen monoxide, NO, reacts with hydrogen, H_2 , according to the following equation:



If the mechanism for this reaction were,



which of the following rate laws would we expect to obtain experimentally?

- A. $\text{Rate} = k[\text{H}_2\text{O}_2][\text{H}_2]$ B. $\text{Rate} = k[\text{NO}]^2[\text{H}_2]$
C. $\text{Rate} = k[\text{NO}]^2[\text{H}_2]^2$ D. $\text{Rate} = k[\text{NO}][\text{H}_2]$
-

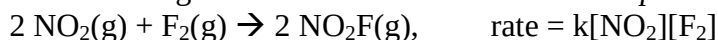
22. Which one of the following is the intermediate?

- A. $\text{NO}(\text{g})$ B. $\text{H}_2\text{O}(\text{g})$ C. $\text{H}_2\text{O}_2(\text{g})$ D. $\text{H}_2(\text{g})$
-

23. The rate constant for a reaction is $1.5 \times 10^{-2} \text{ s}^{-1}$ at 775 K and $3.5 \times 10^{-2} \text{ s}^{-1}$ at 825 K. What is the activation energy? ($R = 8.314 \text{ J/Kmol}$)

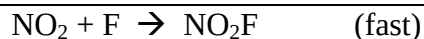
- A. 60 kJ/mol B. 90 kJ/mol C. 120 kJ/mol D. 180 kJ/mol
-

24. The following reaction and rate law have been *experimentally* determined:



Which of the following mechanisms is consistent with these data?

- A. $2\text{NO}_2 + \text{F}_2 \rightarrow 2\text{NO}_2\text{F}$ (one step)
B. $\text{NO}_2 + \text{F}_2 \rightarrow \text{NO}_2\text{F} + \text{F}$ (fast)
 $\text{NO}_2 + \text{F} \rightarrow \text{NO}_2\text{F}$ (slow)
C. $\text{F}_2 \rightarrow 2\text{F}$ (slow)
 $2\text{NO}_2 + 2\text{F} \rightarrow 2\text{NO}_2\text{F}$ (fast)
D. $\text{NO}_2 + \text{F}_2 \rightarrow \text{NO}_2\text{F} + \text{F}$ (slow)
-



25. Nitrogen monoxide, NO, reacts with hydrogen, H₂, to produce nitrogen, N₂, and water, H₂O,
 $2 \text{NO}(\text{g}) + 2 \text{H}_2(\text{g}) \rightarrow \text{N}_2(\text{g}) + 2 \text{H}_2\text{O}(\text{g})$

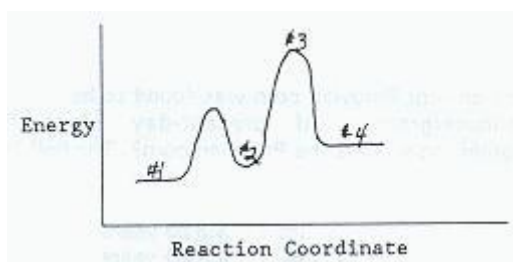
Consider the following mechanism for this reaction:

Step 1:	$2\text{NO}(\text{g}) \rightleftharpoons \text{N}_2\text{O}_2(\text{g})$	fast
Step 2:	$\text{N}_2\text{O}_2(\text{g}) + \text{H}_2(\text{g}) \rightarrow \text{N}_2\text{O}(\text{g}) + \text{H}_2\text{O}(\text{g})$	slow
Step 3:	$\text{N}_2\text{O}(\text{g}) + \text{H}_2(\text{g}) \rightarrow \text{N}_2(\text{g}) + \text{H}_2\text{O}(\text{g})$	fast

of the following rate laws would be consistent with the mechanism proposed above?

- A. Rate = $k[\text{NO}]^2[\text{H}_2]^2$ B. Rate = $k[\text{NO}][\text{H}_2]$ C. Rate = $k[\text{H}_2]^2[\text{N}_2]$
D. Rate = $k[\text{NO}]^2[\text{H}_2]$ E. Rate = $k[\text{NO}]^2$
-

26. With respect to the figure below, which choice correctly identifies all of the numbered positions?



	#1	#2	#3	#4
A.	catalyst	catalyst	activated complex	product
B.	reactants	activated complex	intermediate	product
C.	reactants	Activated complex	catalyst	product
D.	reactants	intermediate	activated complex	Product
E.	reactants	intermediate	activated complex	catalyst

27. The Rate Law for the reaction $\text{NH}_4^+(\text{aq}) + \text{NO}_2^-(\text{aq}) \rightarrow \text{N}_2(\text{g}) + 2 \text{H}_2\text{O}(\text{l})$ is given by rate = $k[\text{NH}_4^+][\text{NO}_2^-]$. The rate constant of the reaction is $3.0 \times 10^{-4} \text{ M}^{-1}\text{s}^{-1}$. The rate of the reaction when $[\text{NH}_4^+] = 0.26 \text{ M}$ and $[\text{NO}_2^-] = 0.080 \text{ M}$ is _____.
A. $6.2 \times 10^{-6} \text{ Ms}^{-1}$ B. $1.3 \times 10^{-4} \text{ Ms}^{-1}$ C. $2.4 \times 10^{-8} \text{ Ms}^{-1}$ D. $3.2 \times 10^{-3} \text{ Ms}^{-1}$
-

28. What is the half-life for the reaction assuming first-order kinetics if 75% of a reactant decomposes in 60 minutes?
A. 120 min B. 15 min C. 90 min D. 30 min
-

CHAPTER 13			
Q.	An.	Q.	A.
1	D	16	A
2	D	17	C
3	B	18	E
4	B	19	D
5	C	20	C
6	B	21	B
7	C	22	C
8	D	23	B
9	C	24	D
10	D	25	D
11	C	26	D
12	D	27	A
13	C	28	D
14	D	29	
15	D	30	